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Towards KPI-Based Health Care Process Improvement

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Abstract

In healthcare organizations, decision making is both a crucial and a challenging task. Therefore, it's necessary to collect all relevant indicators designed for both effective management and process improvement. These indicators need to reflect not only the performance of business process but also the satisfaction of patients toward the ED (emergency department). In this paper, we concentrate on the selection of the relevant performance measures related to health care process and illustrate the usefulness of the retained measures through a real case study in the emergency department of “Farhat Hached” hospital in Sousse (Tunisia). Thus, this work contributes to the improvement of the healthcare process and helps managers to optimize its performance.

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1. Introduction

Recently, business process management (BPM) has become a valuable asset in the healthcare domain [1]. In addition, it is crucial not only to understand the actual situation in the ED and to remodel the business processes, if necessary but also to continue the improvement of healthcare business processes based on comprehensive measurement of organization's performance. However, performing business process analysis in healthcare organizations is particularly difficult due to the highly dynamic, complex, ad-hoc, and multi-disciplinary nature of healthcare processes [2]. The problem above in such a context is how to perform business process (BP) in more effective and efficient way without ignoring patient satisfaction aspect. Despite the difficulties in defining, selecting and monitoring effective indicators, measuring performance is crucial in the current health agenda. Indeed, Key Performance Indicators (KPIs) provide critical information to the organization for monitoring and predicting business performance in accordance with strategic objectives [3]. Hence, selecting KPIs requires a simultaneous consideration of quantitative and also qualitative indicators. For this reason, for quantitative indicators, we are essentially based on the availability of jBPM log to extract all corresponding relevant data. Indeed, the data collected during the business process execution play a key role in improving the overall performance of the business process itself. And for qualitative indicators which are more difficult to measure because they are related to patients experience in the ED, we use a Likert scale to record the level of satisfaction of the patient toward the ED. Another important aspect in order to achieve this improvement is to analyze the relationships between indicators. In our work, this is done using data mining techniques to find all interesting rules.

Some authors [4, 5, 6, 7] have pointed out the vagueness, imprecision of KPI values that they are intended to represent, and the lack of an explicit representation of their semantics. According to [7], the major obstacles for effective design and management of PI monitoring systems are related to the facts that PIs are complex objects with an aggregate/compound nature. This often leads to unawareness of indicator semantics as well as of dependencies among indicators. So, the authors in [7] propose to enrich the data cube model with the formal description of the structure of an indicator given in terms of its algebraic formula and aggregation function. Similarly, in [8], Ortega establishes that, in practice, process performance indicators (PPIs) are defined informally usually in ad-hoc natural language, with its well-known problems or they are defined from an implementation perspective, hardly understandable to not-technical people. In order to solve this problem, the authors propose an approach to improve the definition of PPIs using templates and linguistic patterns.

In our paper, several qualitative and quantitative measurements are retained, and a set of association rules organized for overall process improvement purposes. In this view, to measure the performance of ED, a list of KPIs is gathered from healthcare process and domain expert and is used for testing our proposed approach for BP KPIs selection and modeling. In this work, we are based on our observation, conversation with experts and questionnaire with patients to get all available data to acquire all KPI values. This data will enable us to put together a comprehensive picture of the BP.

The remainder of this paper is organized as follows: section 2 gives an overview of the emergency healthcare process. Section 3 deals with the various corresponding performance measurements. Section 4 focuses on building association rules using data mining techniques, to describe the relationships that hold between different KPIs. The last section gives a brief conclusion and discusses some of our future work.

2. Case Study Scenario

The process considered in this work is the process of healthcare. The main goal of this process is to better satisfy the patients and to offer faster services with better quality. The design phase involves us to look at the big picture of the ED. So, we need to focus on what components are involved in the BP. First, at the beginning of this process, every patient has to go through the Registration activity. After that, in order to arrive at a preliminary conclusion about the status of the patient, sorting activity represents the second task in the BP and the first point of contact with medical staff. This activity consists in recording the preliminary observations and prioritizing the patients, according to their degree of urgency. The following tasks depend on the status of the patient. We find various cases of consultations such as Consultation in the delayed emergency sectors in the case of non-urgent patients, Consultation in the box, it can be a simple consultation or surgical consultation, Consultation in the crash room the patient is of serious harm

which requires immediate medical attention and finally the last case of Consultation in the supervision room if the condition of the patient is not stable.

At the end of this process, three possible cases exist: the patient is treated and leaves the emergency department, the patient is sent to another service (hospitalization or specialized consultation) to ensure continuity of care or the situation of the patient requires long treatments.

The proposed business process model is deployed with the use of jBPM software, where 100 instances have been created through the execution of the process. Our choice of jBPM is motivated by its capability in answering healthcare performance measurement needs, such as the availability of history logs of all process instances as the data source of measurement, the capability of storing logs and information persistence about each performed activity or occurred event that can be exploited. It also provides Business Activity Monitoring (BAM) capabilities such as the Reporting module (also called Dashboard builder). In the next section, we concentrate on research and gathering information about our ED KPIs.

3. Healthcare indicators

There are many measurements that could be used in the evaluation of healthcare process, but after discussion with experts, we arranged a list of relevant indicators. Those listed in tables 1 and 3 are considered to provide the most significant values for decision making. Hence, this step is one of the most important steps because it is based on a deeper understanding of how KPIs are linked to work in particular contexts including who will use them and what decision they support. It involves the following stages:

- Analyze the requirements to formalize healthcare performance indicators. This analysis involves domain experts.
- Extract the possible indicators related to the BP and build other indicators by means of diverse techniques such as inquiries, questionnaires or conversations.
- Establish a preliminary list of the possible indicators including quantitative and qualitative indicators
- Analyze and validate the list of performance measures with the experts.

From this analysis, 32 quantitative indicators and 19 qualitative indicators were selected for this study. The quantitative KPIs are derived from a close look at the activities involved in the BP. Moreover, two specific indicators were added by experts. All KPIs are evaluated and validated through interviews with experts in the health care field. The obtained list of quantitative KPIs is shown in Table 1.

Table 1. Quantitative indicators.

Name of Quantitative KPI
Quanti_KPI1 Number of patients in registration
Quanti_KPI2 Number of patients in sorting
Quanti_KPI3 Number of patients in consultation box
Quanti_KPI4 Number of patients in delayed emergency
Quanti_KPI5 Number of patients in the supervision room
Quanti_KPI6 Number of patients in the crash room
Quanti_KPI7 The duration of registration by patient
Quanti_KPI8 The duration of sorting by patient
Quanti_KPI9 The duration of consultation by patient
Quanti_KPI10 The duration in supervision room by patient
Quanti_KPI11 The duration of delayed emergency by patient
Quanti_KPI12 The duration in crash room by patient

Quanti_KPI13	The duration of all activities by patient
Quanti_KPI14	The min duration of each activity
Quanti_KPI15	The max duration of each activity
Quanti_KPI16	The average duration of each activity
Quanti_KPI17	The waiting time before registration and payment
Quanti_KPI18	The waiting time before sorting
Quanti_KPI19	The waiting time before consultation in box
Quanti_KPI20	The waiting time before consultation in the supervision room
Quanti_KPI21	The waiting time before consultation in crash room
Quanti_KPI22	The waiting time before consultation in delayed emergency
Quanti_KPI23	The waiting time per patient in all activities
Quanti_KPI24	The min waiting time for all patients in all activities
Quanti_KPI25	The max waiting time for all patients in all activities
Quanti_KPI26	The average waiting time for all patients in all activities
Quanti_KPI27	The total time spent in the Emergency Department by the patient in all activities
Quanti_KPI28	The max of total time spent in the ED by all patients in all activities
Quanti_KPI29	The min of total time spent in the ED by all patients in all activities
Quanti_KPI30	The average of total time spent in the ED by all patients in all activities
Quanti_KPI31	Number of doctors by date
Quanti_KPI32	Number of nurses by date

In the first step, we analyzed the relationship among history logs to see how the log tables are related to each other and to identify what information can be used for performance measurement purpose.

These log information represent an input to build reports in Business dashboard module. In the dashboard, the data can be retrieved from both relational jBPM databases using SQL query and CSV files. Measuring performance indicators is basically based on queries to the data source (by retrieving the query results). For example, Figure 1 represents the dashboards in KIE Workbench environment related to task duration details.

process_id	start_date	end_date	taskname	createddate	enddate	status	task dura
211	05/21/16 10:50	05/21/16 11:50	Consultation Del	05/21/16 11:29	05/21/16 11:50	Completed	0h 5m 25s
212	05/21/16 10:28	05/21/16 11:15	Registration	05/21/16 10:28	05/21/16 10:32	Completed	0h 1m 40s
212	05/21/16 10:28	05/21/16 11:15	Sorting	05/21/16 10:32	05/21/16 10:46	Completed	0h 3m 12s
212	05/21/16 10:28	05/21/16 11:15	Consultation Del	05/21/16 10:46	05/21/16 11:15	Completed	0h 5m 11s
448	05/26/16 11:35	05/26/16 5:35 P	Registration	05/26/16 11:35	05/26/16 11:37	Completed	0h 1m 40s
213	05/22/16 11:45	05/22/16 12:08	Registration	05/22/16 11:45	05/22/16 11:46	Completed	0h 1m 10s
213	05/22/16 11:45	05/22/16 12:08	Sorting	05/22/16 11:46	05/22/16 11:50	Completed	0h 2m 20s
213	05/22/16 11:45	05/22/16 12:08	Consultation in b	05/22/16 11:50	05/22/16 12:08	Completed	0h 18m 32s
214	05/22/16 11:42	05/22/16 12:17	Registration	05/22/16 11:42	05/22/16 11:43	Completed	0h 1m 8s
214	05/22/16 11:42	05/22/16 12:17	Sorting	05/22/16 11:43	05/22/16 11:46	Completed	0h 1m 3s

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Fig. 1. Task duration details

From the available data in jBPM logs, we conclude that not all KPI values can be directly retrieved from these logs so we create another data table which contains all real values related to our research. An overview of this table is presented in table 2.

Table 2. An example for real value from quantitative indicators table

	quanti_kpi7 character varying(255)	quanti_kpi8 character varying(255)	quanti_kpi9 character varying(255)	quanti_kpi10 character varying(255)	quanti_kpi11 character varying(255)	quanti_kpi12 character varying(255)	quanti_kpi13 character varying(255)
22	00:00:18	00:01:22	00:04:17	00:01:20			00:07:18
23	00:01:08	00:00:53	00:05:14				00:07:16
24	00:03:13	00:00:10	00:01:26				00:04:50
25	00:01:04	00:02:16			00:02:10		00:05:31
26	00:01:16	00:01:33	00:24:31				00:27:21
27	00:02:11	00:02:59	01:35:20				01:40:31
28	00:02:11	00:03:23	00:15:17	02:10:17			02:31:09
29	00:03:11	00:03:13			00:08:42		00:15:07
30	00:00:37	00:03:57	00:01:11	01:50:20			01:56:06

For qualitative indicators, a qualitative inquiry is necessary to get close enough to the patients and capture the level of their satisfaction and to examine the quality of care provided to patients attending the ED. Hence, at the end of the process, each patient was invited to provide feedback about his satisfaction. The qualitative indicators which are in touch with the satisfaction towards the ED were synthesized in table 3.

Table 3. List of qualitative indicators

Name of Qualitative KPI
Quali_KPI1 interest and attention brought by paramedical staff
Quali_KPI2 interest and attention brought by medical staff
Quali_KPI3 paramedical staff availability
Quali_KPI4 medical staff availability
Quali_KPI5 medical equipment availability
Quali_KPI6 hospital staff is well dressed
Quali_KPI7 the quality of care for patients by paramedical staff
Quali_KPI8 the quality of care for patients by medical staff
Quali_KPI9 the clarity of information
Quali_KPI10 installation in the waiting room before delayed emergency
Quali_KPI11 installation in the waiting room before registration
Quali_KPI12 installation in the waiting room before consultation in box room
Quali_KPI13 installation in the sorting room
Quali_KPI14 installation in the supervision room
Quali_KPI15 installation in the crash room
Quali_KPI16 the overall waiting time before treatment by paramedical personnel
Quali_KPI17 the overall waiting time before treatment by medical staff
Quali_KPI18 the regularity of doctor visits in the supervision room
Quali_KPI19 the regularity of doctor visits in the crash room

Those indicators concern the qualitative aspects of care in the ED, such as staff attitudes towards patients, but they also concern some quantitative aspects such as paramedical staff availability and the overall waiting time before treatment by paramedical personnel. This is due for example to the fact that sometimes the nurse is available but the

patients who are in less urgent state are not given sufficient attention by nursing staff. So, in order to provide a higher level of quality and consistency with quantitative measurement and with the processes in place, we record the patient's level of satisfaction with those indicators. The aim of this questionnaire is to determine why patients are unsatisfied and what we can do to make them better satisfied.

In order to fulfill the qualitative database, we were based on the responses of the patients, and consequently, a set of qualitative results is recorded by using a Likert scale. To do that, we created a new table in the database with the respective responses of the patients as the values of KPIs. Since the exactly followed paths in the process are different, the number of questions asked to each specific patient varies and then some columns in the database have a null value. Table 4 shows some data value in the qualitative table.

Table 4. An example for real value from qualitative indicators table

processinstanceid [PK] bigint	quali_kpi1 character varying(255)	quali_kpi2 character varying(255)	quali_kpi3 character varying(255)	quali_kpi4 character varying(255)	quali_kpi5 character varying(255)	quali_kpi6 character varying(255)	quali_kpi7 character varying(255)	quali_kpi8 character varying(255)
168	satisfied	satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied
170	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied
172	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied
173	satisfied	satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied
174	satisfied	satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied
175	dissatisfied	very dissatisfied	very satisfied	very satisfied	very satisfied	very satisfied	dissatisfied	dissatisfied
177	satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied
178	very satisfied	very satisfied	very satisfied	very satisfied	dissatisfied	very satisfied	very satisfied	very satisfied
180	very satisfied	very satisfied	dissatisfied	dissatisfied	very satisfied	very satisfied	very satisfied	very satisfied
181	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied	very satisfied		

At this stage, we track the progress of this process and we identify and extract data about the KPIs. Data collected during business process execution are used for deriving the key performance indicators (KPI) that allow continuous tracking of the process behavior and measurement of process-specific goals. In addition, we record all real values of indicators of both quantitative and qualitative aspect which offer a better comprehension of the situation. But in order to give more sense and to ensure that healthcare process is fulfilling the needs and expectations of the ED and its patients, this data need to be checked and evaluated by the organization to detect if KPI values are reaching the desired results. As a result, we create another table which contains all estimated values related to the patient instance. This table forms the basis for continuous process optimization.

4. Data mining technique

Furthermore, it is crucial to discover interesting relations between KPIs. In fact, to make any decision, it is important to analyze all the relevant row data of our performance measurements. For this purpose, we use data mining, especially association rule learning as a research method in this stage. An association rule is defined as an implication expression which contains antecedent and consequent, where antecedent and consequent are item sets. In our use case, an Item set is a collection of one or more indicators. By using Apriori algorithm we aim to find frequent associations and correlations among sets of items.

Based on Estimated KPI value table, we designed associated relation. This table contains two possible values “ok” and “Not ok”. After comparing the real value and the range of tolerance for each of KPIs; we decided to add “ok” value when the value of this KPI is appropriate and accepted by the ED and “Not Ok” value if the KPI real value in not acceptable (under the range of tolerance). As aforesaid, this table contains 100 instance records which can be statistically considered as significant. Association rule mining algorithm needs to be configured before learning. So, we give appropriate values for the parameters in advance in order to obtain a good number of rules. The result obtained in Figures 2 and 3 are based on Apriori algorithm to predict the occurrence of an item based on the occurrences of other items in the transaction.

Id	Antecedent	Consequent	Length	Support	Confidence	Recall	F-measure	Lift	Conviction
1	quali_kpi17=ok	quali_kpi16=ok	2	0.6200	0.9118	0.9841	0.9466	1.4472	3.7255
2	quali_kpi16=ok	quali_kpi17=ok	2	0.6200	0.9841	0.9118	0.9466	1.4472	10.5147

Fig. 2. Example 1 of association rules

For example, for Rule 1 (Id=1) the fraction of transactions that contains both quali_KPI17 (The overall waiting time before treatment by medical staff) and quali_KPI16 (The overall waiting time before treatment by paramedical

5	quali_kpi1=ok	quali_kpi2=ok	2	0.6700	0.9853	0.9054	0.9437	1.3315	9.2647
6	quali_kpi3=ok	quali_kpi2=ok	2	0.6600	0.8800	0.8919	0.8859	1.1892	2.0382
7	quali_kpi4=ok	quali_kpi2=ok	2	0.6700	0.8816	0.9054	0.8933	1.1913	2.0647
8	quali_kpi5=ok	quali_kpi2=ok	2	0.6300	0.8400	0.8514	0.8456	1.1351	1.5679
9	quali_kpi6=ok	quali_kpi2=ok	2	0.7100	0.8161	0.9595	0.8820	1.1028	1.3858
10	quali_kpi7=ok	quali_kpi2=ok	2	0.6200	0.9118	0.8378	0.8732	1.2321	2.6471
11	quali_kpi8=ok	quali_kpi2=ok	2	0.6400	0.9143	0.8649	0.8889	1.2355	2.7227

personnel) is 0.62 and the confidence for this rule is 0.81. This value measures how often items in quali_KPI16 appear in transactions that contain quali_KPI17.

We can see in figure 3 that quali_KPI2 (interest and attention brought by medical staff) is consequent in the rule form, which can be used to determine what another measurement should be associated with in order to have a high level of satisfaction.

quali_KPI1 interest and attention brought by paramedical staff, quali_KPI3 paramedical staff availability,

Fig. 3. Example 2 of association rules

quali_KPI4 medical staff availability, quali_KPI5 medical equipment availability, quali_KPI6 hospital staff are well dressed, quali_KPI7 the quality of care for patients by paramedical staff and quali_KPI8 the quality of care for patients by medical staff are the antecedent. Those rules can be used to see what other KPIs should be taken into account to promote a high satisfaction with the quali_KPI2.

This analysis of frequent items aims to find all interesting rules that correlate the presence of one set of items with that of another set of items.

4. Conclusion

Performance measures enable ED to carry out a diagnosis for determining strengths and weaknesses of BP, which plays a central role in the business process. In this paper, we explored several KPIs using jBPM management features. The key performance indicators offer the ability to evaluate the BP using jBPM dashboards, which provide a snapshot of KPIs of particular interest to internal (for example doctors and nurses) and external (patient) stakeholders that focus on their capacity to provide care in a way that responds to the basic questions posed by patients. After collecting quantitative and qualitative data, department care teams will identify the most important KPIs, measure them, report the results, and improve them as needed like improving patient flow and patient satisfaction degree. Moreover, we analyzed some association rules related to real value instances in the healthcare BP. As part of our future work, we plan to enrich our KPI knowledge using the semantic technique to present all relationships between KPIs or between activities and KPIs.

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